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REVIEW ARTICLE

Tetrahydrobiopterin: biochemistry and pathophysiology

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BH_4 (*6R*-L-erythro-5,6,7,8-tetrahydrobiopterin) is an essential cofactor of a set of enzymes that are of central metabolic importance, including four aromatic amino acid hydroxylases, alkylglycerol mono-oxygenase and three NOS (NO synthase) isoenzymes. Consequently, BH_4 is present in probably every cell or tissue of higher organisms and plays a key role in a number of biological processes and pathological states associated with monoamine neurotransmitter formation, cardiovascular and endothelial dysfunction, the immune response and pain sensitivity. BH_4 is formed *de novo* from GTP via a sequence of three enzymatic steps carried out by GTP cyclohydrolase I, 6-pyruvoyltetrahydropterin synthase and sepiapterin reductase. An alternative or salvage pathway involves dihydrofolate reductase and may play an essential role in peripheral tissues. Cofactor regeneration requires pterin-4a-carbinolamine dehydratase and dihydropteridine reductase, except for NOSs, in which the BH_4 cofactor undergoes a one-electron redox cycle without the need for additional regeneration enzymes. With regard to the regulation of cofactor biosynthesis, the major controlling point is GTP cyclohydrolase I. BH_4 biosynthesis is controlled in mammals by hormones and cytokines. BH_4 deficiency due to autosomal recessive mutations in all enzymes, except for sepiapterin reductase, has been described as a cause of hyperphenylalaninaemia. A major contributor to vascular dysfunction associated with hypertension, ischaemic reperfusion injury, diabetes and others, appears to be an effect of oxidized BH_4, which leads to an increased formation of oxygen-derived radicals instead of NO by decoupled NOS. Furthermore, several neurological diseases have been suggested to be a consequence of restricted cofactor availability, and oral cofactor replacement therapy to stabilize mutant phenylalanine hydroxylase in the BH_4-responsive type of hyperphenylalaninaemia has an advantageous effect on pathological phenylalanine levels in patients.

Key words: alkylglycerol mono-oxygenase (AGMO), aromatic amino acid hydroxylase, GTP cyclohydrolase I (GTPCH), NO synthase (NOS), 6-pyruvoyltetrahydropterin synthase (PTPS), sepiapterin reductase (SR).

INTRODUCTION

In the time period following publication of our last version of this review article on the mammalian aspects of BH_4 (*6R*-L-erythro-5,6,7,8-tetrahydrobiopterin) biosynthesis, regeneration and functions in 2000 [1], approximately 500 research publications and more than ten overview articles have been published in the last 10 years according to a PubMed search using the search term 'tetrahydrobiopterin', reflecting an inflation of interest in this topic. The attention in the field has shifted from structural and metabolic-molecular interest to homoeostasis of the BH_4 system and pathophysiology of BH_4 deficiency, with a greater emphasis on the *in vivo* effects on NOS (NO synthase) decoupling. It is not the idea of this update to again summarize the topics on reaction mechanisms and enzyme structures etc., knowledge of which has not changed significantly since 2000, but to build on previous facts and to summarize the more recent activities in the field. However, we have not included the structural analyses of the BH_4-dependent aromatic amino acid hydroxylases, which have been solved and reviewed elsewhere [2,3]. The 'news headlines' are the discovery of human SR (sepiapterin reductase) deficiency concomitant unexpectedly with central monoamine neurotransmitter deficiency but normal plasma phenylalanine levels, the description of two new BH_4-dependent enzymes, AGMO (alkylglycerol mono-oxygenase) and the brain-specific tryptophan hydroxylase isomer TPH2, and the BH_4-responsive type of hyperphenylalaninaemia where oral BH_4 supplementation stabilizes mutant PAH (phenylalanine hydroxylase) in patients. Furthermore, we summarize the almost exploding research activities on the mechanism of NOS, the radical function of BH_4, and the potential consequences of BH_4 decoupling from NOS. With regard to *in vivo* gene function studies and modelling human disorders, mouse models have become a vital tool in BH_4 research too. This aspect has been taken into account particularly by adding a section on mouse models.

BIOSYNTHESIS AND REGENERATION OF BH_4

An overview on the human BH_4 metabolic and all BH_4 cofactor-dependent enzymes is given in Table 1, which includes six proteins or enzymes involved in the biosynthesis, regeneration and regulation of BH_4, as well as all eight BH_4 cofactor-dependent enzymes known to date. The reactions catalysed by the BH_4-dependent enzymes are summarized in Figure 1.

Abbreviations used: AGMO, alkylglycerol mono-oxygenase; ApoE, apolipoprotein E; AR, aldose reductase; BH_4, *6R*-L-erythro-5,6,7,8-tetrahydrobiopterin; CR, carbonyl reductase; DCoH1, dimerization cofactor of hepatocyte nuclear factor 1α; DHFR, dihydrofolate reductase; DHPR, dihydropteridine reductase; DRD, dopa-responsive dystonia; GFRP, GTP cyclohydrolase I feedback regulatory protein; GTPCH, GTP cyclohydrolase I; HNF-1α, hepatocyte nuclear factor 1α; IGF-1, insulin-like growth factor-1; ki, knockin; ko, knockout; NOS, NO synthase; PAH, phenylalanine hydroxylase; PCD, pterin-4a-carbinolamine dehydratase; PTPS, 6-pyruvoyltetrahydropterin synthase; SR, sepiapterin reductase; TH, tyrosine hydroxylase; TPH, tryptophan hydroxylase.

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Table 1 Human BH_4 metabolic and BH_4 cofactor-dependent enzymes

AKR1B1, aldose reductase 1B1; AKR1C3, 3α-hydroxysteroid dehydrogenase type 3; CBR1, carbonyl reductase 1; eNOS, endothelial NOS; iNOS, inducible NOS; nNOS, neuronal NOS; OMIM, Online Mendelian Inheritance in Man.

Enzyme	EC number	Gene symbol	OMIM number	Chromosome location	Number of exons (amino acids)	Size (kDa × number of subunits)	Mouse model
GTPCH	3.5.4.16	*GCH1*	600225	14q22.1-q22.2	6 (250)	27.9 × 10	Not available *hph-1* [177–181]
GFRP (P35)	None	*GCHFR*	602437	15q15	3 (84)	9.7 × 5	*Gchfr* conditional ko†
PTPS	4.2.3.12*	*PTS*	612719	11q22.3-q23.3	6 (145)	16.4 × 6	*Pts*-ko [182,183]/*Pts*-ki [184]
SR	1.1.1.153	*SPR*	182125	2p13	3 (261)	28.0 × 2	*Spr*-ko [185]
CR (CBR1)	1.1.1.184	*CBR1*	114830	21q22.13	3 (277)	30.4 × 2	Not available
AR (AKR1B1)	1.1.1.21	*AKR1B1*	103880	7q35	10 (316)	35.9 × 2	*Akr1b1*-ko [193]
AKR1C3	1.1.1.213	*AKR1C3*	603966	10p15-p14	9 (323)	36.9 × 2	Not available
DHFR	1.5.1.3	*DHFR*	126060	5q11.2-q13.2	6 (187)	21.5 × 2	Not available
PCD/DCoH1	4.2.1.96	*PCBD1*	126090	10q22	4 (103‡)	11.9 × 4§	*DCoH*-ko [187]
DHPR	1.6.99.7	*QDPR*	612676	4p15.3	7 (244)	25.8 × 2	*Qdpr*-ko [189]
PAH	1.14.16.1	*PAH*	612349	12q24.1	4 (452)	51.9 × 2 or × 4	*enu1*/*enu2* [194,195]
TH (TYH)	1.14.16.2	*TH*	191290	11p15.5	14 (TH-4; 528)‖	58.5 × 4	*Th*-ko [196,197]
TPH1 (TPRH, TRPH)	1.14.16.4	*TPH1*	191060	11p15.3-p14	11 (444)	51.0 × 4	*Tph1*-ko [198]
TPH2 (NTPH)	1.14.16.4	*TPH2*	607478	12q15	11 (490)	65.0 × 4	*Tph2*-(conditional) ko [199–201]
AGMO (TMEM195)	1.14.16.5	*AGMO*	613738	7p21.2	13 (445)	51.5 × ?	*Agmo* conditional ko¶
NOS1 (nNOS, NOSI)	1.14.19.39	*NOS1*	163731	12q24.2-q24.31	28 (1434)	161.0 × 2	*nNos*-ko [205]
NOS2 (iNOS, mNOS, NOSII)	1.14.19.39	*NOS2*	163730	17cen-q11.2	27 (1153)	131.1 × 2	*iNos*-ko [205]
NOS3 (eNOS, ECNOS, NOSIII)	1.14.19.39	*NOS3*	163729	7q36	26 (1203)	133.3 × 2	*eNos*-ko [205]

*EC 4.6.1.10 created in 1999; deleted in 2000.

†D. Adamsen, T. Scherer, N. Blau and B. Thöny, unpublished work.

‡In liver, the protein was found to lack the starting methionine residue.

§Homotetrameric (α_4) only as a carbinolamine dehydratase, heterotetrameric ($\alpha_2\beta_2$) in complex with HNF-1α.

‖For TH, four splice variants in intron 1 exist, TH-1 (13 exons, 497 amino acids), TH-2 (13 exons, 501 amino acids), TH-3 (14 exons, 524 amino acids) and TH-4 (14 exons, 528 amino acids) [227].

¶K. Watschinger, N. Yannoutsos, M.A. Keller, G. Golderer, G. Werner-Felmayer and E.R. Werner, unpublished work.

Biosynthesis

BH_4 biosynthesis proceeds in a *de novo* pathway in a Mg^{2+}-, Zn^{2+}- and NADPH-dependent reaction from GTP via two intermediates, 7,8-dihydroneopterin triphosphate and 6-pyruvoyl-5,6,7,8-tetrahydropterin (Figure 2). The three enzymes GTPCH (GTP cyclohydrolase I), PTPS (6-pyruvoyltetrahydropterin synthase) and SR are required to carry out the proper stereospecific reaction to make BH_4. Using the crystallographic structures, including the characteristics of the active centres of all three enzymes, the essential information for the interpretation of the reaction mechanism is established. Moreover, NMR studies on the reaction mechanisms of all three enzymes revealed the details of the hydrogen-transfer process and the stereochemical course of the reactions [4].

The initial step is carried out by GTPCH, a homodecamer consisting of a tightly associated dimer of two pentamers [5]. GTPCH contains ten equivalent active centres with 10-Å (1 Å = 0.1 nm)-deep pockets. The interface of three subunits, two from one pentamer and one from the other, forms an active site. The atomic structure of this pocket is not only highly selective for GTP, but also provides residues for complete charge compensation in order to render obsolete Mg^{2+}-assisted binding to the protein, as found in other nucleoside-triphosphate-binding proteins. A catalytic mechanism was proposed on the basis of structural analysis obtained from *Escherichia coli* GTPCH co-crystals with the dGTP analogue and several active-site mutants [6]. This was later shown to be aided by an essential zinc (Zn^{2+}) in the active site of human and bacterial GTPCH [7].

The reaction from 7,8-dihydroneopterin triphosphate to 6-pyruvoyltetrahydropterin is catalysed by PTPS in a Mg^{2+}- and Zn^{2+}-dependent reaction without consuming an external reducing agent (Figure 2). This conversion involves a stereospecific reduction by an internal redox transfer between atoms N-5, C-6 and C-1′, oxidation of both side-chain hydroxy groups, and an unusual triphosphate elimination at the C-2′–C-3′ bond in the side chain. Crystallographic analysis revealed that PTPS is composed of a pair of trimers arranged in a head-to-head fashion to form the functional hexamer [8]. The homohexamer contains six active sites that are located on the interface of three monomers, two subunits from one trimer and one subunit from the other trimer. The catalytic centre and the reaction mechanism were studied by crystallographic and kinetic analysis of the rat wild-type and mutant PTPS [9]. Each catalytic centre harbours a Zn^{2+}-metal binding site in a 12-Å-deep cavity. The active-site pocket contains an additional two catalytic motifs: a Zn^{2+}-binding site and an intersubunit catalytic triad formed by a cysteine, an aspartate and a histidine residue. The tetravalent co-ordination of the transition metal is accomplished via the Nε-atoms of three histidine residues and a fourth ligand provided by the pyruvoyl moiety of the 7,8-dihydroneopterin triphosphate substrate [10].

The final step is the NADPH-dependent reduction of the two side-chain keto groups of 6-pyruvoyltetrahydropterin by SR (Figures 2 and 3). The overall structure of SR is a homodimer stabilized by a common four-helix bundle [11]. Each monomer contributes with two α-helices to the central dimerization domain and forms a separate complex composed of seven parallel β-sheets surrounded by α-helices. On the

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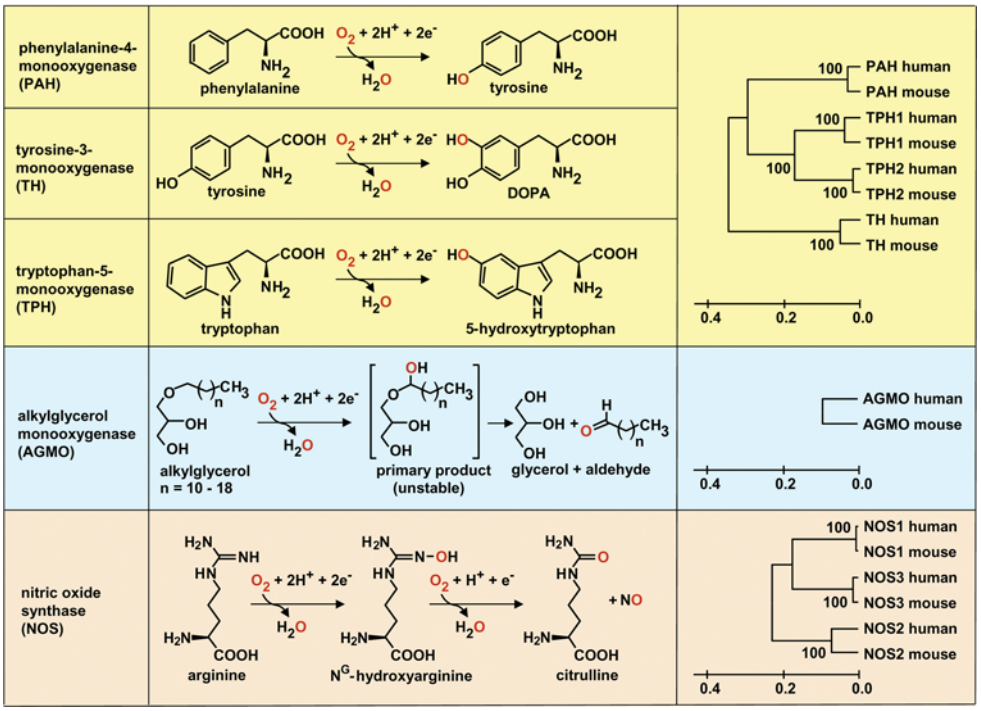

Figure 1 Overview of BH_4 cofactor-dependent enzyme reactions

All known BH_4-dependent reactions are mixed function mono-oxygenases, i.e. one oxygen atom is incorporated into the substrate and one oxygen atom is reduced to water (shown in red). Sequence analysis groups BH_4-dependent enzymes to three separate clusters: the four aromatic amino acid hydroxylases (PAH, TH, TPH1 and TPH2; yellow background), AGMO (blue background), and the three NOSs (pink background) as shown in the phylogenetic trees of the protein sequences from human and mouse in the right-hand part of the Figure. There is no sequence homology between these three groups. The numbers shown in the phylogenetic trees indicate the probability in percentage of the branching at this point as measured by a bootstrap analysis using UPGMA (un-weighted pair group method using arithmetic averages) [226]. The numbers in the horizontal axis indicate the average portion of amino acids substituted. Note that BH_4 plays a different role for the aromatic amino acid hydroxylases and AGMO in comparison with the NOSs: whereas in the first group, BH_4 is activating the molecular oxygen and is externally regenerated (for details, see the text and Figure 4), for the NOSs, the cofactor supplies an electron for each of the two reactions from arginine to citrulline and NO via N^G-hydroxyarginine. BH_4 is regenerated when bound to the enzyme (for further details, see the text and Figure 5).

basis of kinetic, crystallographic and NMR data, the initial step is a NADPH-dependent reduction at the side-chain C-1′-keto function, leading to the formation of 1′-hydroxy-2′-oxopropyl-tetrahydropterin [4]. Internal rearrangement of the keto group via side-chain isomerization leads to the 1′-keto compound 1′-oxo-2′-hydroxypropyl-tetrahydropterin (also called 6-lactoyl-tetrahydropterin). The 1′-keto compound 6-lactoyl-tetrahydropterin is then reduced to BH_4 in a second NADPH-dependent reduction step. Although the pterin substrate remains bound to the active site, the redox cofactor has to be renewed after the first reduction. It is thus assumed that NADP is exchanged at the opening located at the opposite side of the pterin-binding and entry pocket.

Salvage and alternative pathways

Besides the involvement in the *de novo* biosynthesis of BH_4, SR may also participate in the pterin salvage pathway (Figure 3) by catalysing the conversion of sepiapterin into 7,8-dihydrobiopterin, which is then transformed into BH_4 by DHFR (dihydrofolate reductase) [12].

Although SR is sufficient for completion of BH_4 biosynthesis, other enzymes share the same substrates with SR and are able to catalyse the same reactions: AR (aldose reductase; AKR1B1) [13,14] and CR (carbonyl reductase; CBR1) [15] both participate in the di-keto reduction of the carbonyl side chain of 6-pyruvoyltetrahydropterin *in vivo* (Figure 3). Previously, it was proposed that, in the absence of SR activity, a salvage pathway may alternatively synthesize BH_4 [15]: AR and CR convert 6-pyruvoyltetrahydropterin into 1′-oxo-2′-hydroxypropyl-tetrahydropterin, which is non-enzymatically converted into sepiapterin; CR then reduces sepiapterin to 7,8-dihydrobiopterin, which is reduced to BH_4 by DHFR [16] (Figure 3). Another enzyme of the family of short-chain aldo-keto reductases has been identified, which actively participates in an alternative pathway for BH_4 synthesis [17]. According to the most recent model, BH_4 can be efficiently synthesized, in the absence of SR, by the concerted action of 3α-hydroxysteroid dehydrogenase type 3 (AKR1C3) and AR through the intermediate 1′-hydroxy-2′-oxopropyl-tetrahydropterin; this reaction seems to be much more efficient than the salvage pathway [17].

Regulation of BH_4 biosynthesis

As mentioned above, BH_4 is synthesized via the *de novo* pathway from GTP where GTPCH is the rate-controlling enzyme (see Figure 2). Many details on the regulation of GTPCH, and also GFRP (GTPCH feedback regulatory protein), PTPS and SR were summarized by different authors in previous reviews [1,18,19]. Under conditions of immune stimulation of macrophages by various cytokines or inflammatory mediators, *GCH1* expression and thus GTPCH activity is strongly up-regulated. Additionally, PTPS becomes limiting, as the dephosphorylated and oxidized intermediate neopterin is significantly elevated and accumulated in human plasma, although this is not observed in rodents. Up-regulation of *GCH1* mRNA [together with *TH* (tyrosine

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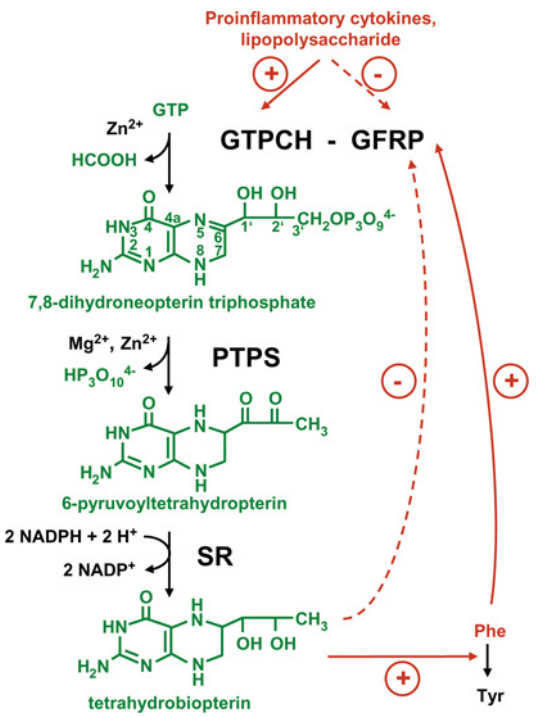

Figure 2 BH_4 *de novo* biosynthesis and regulation

GTPCH is the key regulatory point of BH_4 biosynthesis, at least in higher animals. The expression of the enzyme is regulated by cytokines and hormones. Its enzymatic activity is regulated by GFRP in a BH_4- and phenylalanine-dependent manner. GFRP is down-regulated by pro-inflammatory stimuli which up-regulate GTPCH, possibly to dissociate BH_4 biosynthesis in inflammation from the metabolic control by BH_4/phenylalanine which under normal conditions assures optimal phenylalanine degradation.

hydroxylase) mRNA] in various catecholaminergic brain regions of the mouse was also reported by administration of oestradiol or lipopolysaccharides [20,21]. Besides the transcriptional stimulation of *GCH1*, post-translational regulation of GTPCH involves phosphorylation and enzyme activity modulation via GFRP via the substrate GTP, the pathway end-product BH_4 and (blood) phenylalanine, but not by other amino acids, such as tyrosine or tryptophan. On the basis of the observed regulatory interaction between GTPCH and TH in the fly, it was postulated that an analogous modulation of GTPCH activity via TH may exist in mammalian dopaminergic neurons [22]. The intracellular level of GTP modulates GTPCH activity by co-operative binding and thereby changing the enzyme kinetics. BH_4 and phenylalanine modulate enzymatic activity via GFRP, which binds to GTPCH, thus inducing an as yet unknown conformational change. A consequence of (hepatic) GFRP action is the high plasma BH_4 concentration observed in patients with hyperphenylalaninaemia caused by PAH deficiency [23]. On the other hand, bacterial lipopolysaccharide and H_2O_2 down-regulated expression of GFRP, at least in cultured cells and in rat tissues, rendering BH_4 synthesis independent of metabolic control by phenylalanine [24–26]. Recombinant over-expression of GFRP in a murine endothelial cell line did not alter basal BH_4 synthesis, but significantly attenuated cytokine plus lipopolysaccharide-induced increases in intracellular BH_4 [27]. Down-regulation of GFRP in cultured human aortic endothelial cells increased BH_4 concentrations and GTPCH phosphorylation [28]. In a murine endothelial cell line overexpressing GTPCH, however, modulation of GFRP expression did not alter BH_4 levels [29]. *GCHFR* mRNA studies by Northern blot analysis and *in situ* hybridization revealed that the expression pattern in rat tissues correlates with that of GTPCH, i.e. GFRP is expressed in peripheral organs such as liver and heart, and also in the brain [30–32]. GTPCH has been shown under *in vitro* conditions to exhibit a unique form of allosteric regulation in the liver by direct interaction with GFRP in the presence of phenylalanine or BH_4. Regulation of BH_4 biosynthesis in brain is less clear. GFRP appears to be specifically expressed in some brain areas or cells types, and indirect evidence from studies with primary neuronal cultures indicated that GFRP may regulate BH_4 biosynthesis in serotonin, but not in dopamine neurons [31]. In any case, *in vivo* studies are required to investigate the function of GFPR in brain or other organs.

Evidence for transcriptional control of *PTS* gene expression was observed in human monocytes, where low BH_4 biosynthesis was investigated and shown to be caused by the skipping of exon 3, a major controlling mechanism for PTPS protein activity. The ratio of exon-3-containing to exon-3-lacking *PTS* mRNA due to skipping of exon 3 was not affected by differential mRNA stability or nonsense-mediated mRNA decay, but rather produced less PTPS protein due to a premature stop codon. The observation that exon 3-lacking mRNA was found in diverse cells or tissues, including dermal fibroblasts and brain, further substantiates the hypothesis that skipping of exon 3 could also cause low PTPS activity in other cells or tissues, and is thus a regulating mechanism for enzyme activity [33].

Regeneration

During the catalytic event of aromatic amino acid hydroxylation, molecular oxygen is transferred to the corresponding amino acid and BH_4 is oxidized to BH_4-4a-carbinolamine (also termed 4a-hydroxy-tetrahydrobiopterin; Figure 4). Two enzymes are involved in its subsequent dehydration and reduction to BH_4: PCD (pterin-4a-carbinolamine dehydratase) and DHPR (dihydropteridine reductase). Enzymatic regeneration of BH_4 is essential for phenylalanine metabolism: (i) to ensure a continuous supply of reduced cofactor; and (ii) to prevent accumulation of harmful metabolites produced by rearrangement of BH_4-4a-carbinolamine. The primary structure of PCD is identical with a protein of the cell nucleus, named DCoH1 [dimerization cofactor of HNF-1α (hepatocyte nuclear factor 1α)], which was reported to have a general transcriptional function [34,35]. From here on, PCD will be designated as PCD/DCoH1 protein.

Dehydration of BH_4-4a-carbinolamine, the first product of the reaction of aromatic amino acid hydroxylases (Figure 4), is catalysed by the enzyme PCD/DCoH1. The human cytoplasmic PCD/DCoH1, whose sequence is identical with the rat protein, is a homotetramer with a molecular mass of 11.9 kDa per subunit [36,37]. Using chemically synthesized pterin-4a-carbinolamine, it has been shown that the enzyme displays little sensitivity to the structure or configuration of the 6-substituent of its substrate, and to the 4a(*R*)- and 4a(*S*)-hydroxy stereoisomers [38]. Detailed enzymatic studies on the stereospecificity and catalytic function revealed a dehydration mechanism in which the three histidine residues in PCD/DCoH1 are crucial for activity [39]. The quinoid dihydrobiopterin product is a strong inhibitor of PCD/DCoH1 with a K_i value of approximately one half of its respective K_m value, and no inhibition was observed with 7,8-dihydrobiopterin [38]. Furthermore, PAH is not inhibited by its cofactor product, BH_4-4a-carbinolamine, but rather by primapterin, which is an isomer of biopterin carrying the dihydroxypropyl side chain at position 7 instead of position 6 of the pteridine ring. In the absence of PCD/DCoH1, dehydration of BH_4-4a-carbinolamine

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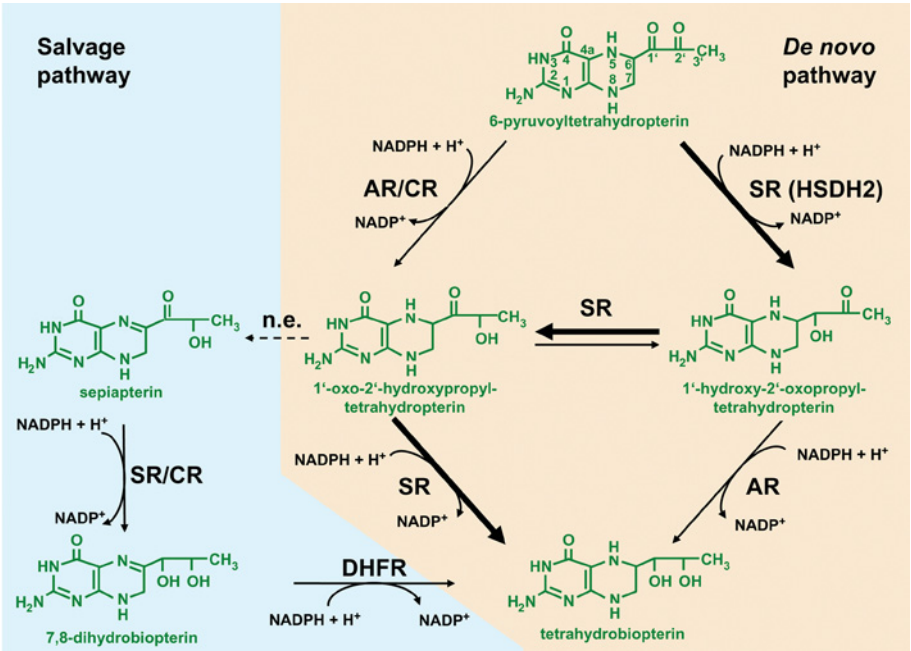

Figure 3 Alternative pathways for the biosynthesis of BH_4

The last two-step reduction of 6-pyruvoyltetrahydropterin to BH_4 in the *de novo* pathway is catalysed by SR. 1′-Hydroxy-2′-oxopropyl- and 1′-oxo-2′-hydroxypropyl-tetrahydropterin are proposed intermediates, the latter is converted non-enzymatically (n.e.) into sepiapterin in the absence of SR (see also Figure 6). AR and CR catalyse the conversion of 6-pyruvoyltetrahydropterin into the 1′-oxo-2′-hydroxypropyl intermediate, and 3α-hydroxysteroid dehydrogenase type 2 (HSDH2) reduces 6-pyruvoyltetrahydropterin to 1′-hydroxy-2′-oxopropyl-tetrahydropterin. In addition, BH_4 can be formed by the reduction of 1′-hydroxy-2′-oxopropyl-tetrahydropterin through AR, however, at a much lower rate. In the salvage pathway, the final step of reduction of 7,8-dihydrobiopterin to BH_4 is catalysed by the methotrexate-sensitive enzyme DHFR.

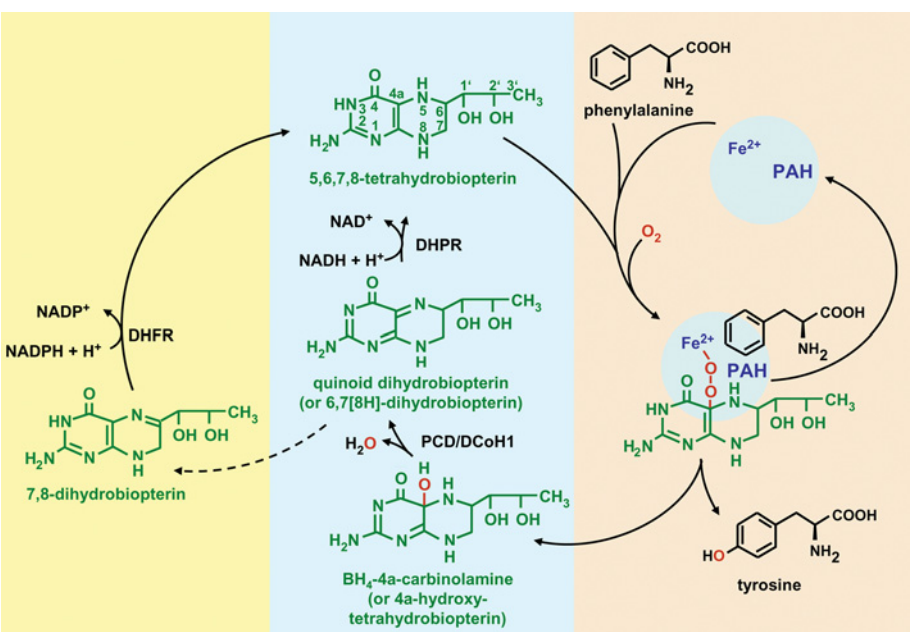

Figure 4 Simplified scheme of the PAH reaction

PAH contains a catalytically active non-haem iron. BH_4 (green) is positioned close to this iron atom and participates in activation of the oxygen (red) in forming a presumed peroxo intermediate. Upon hydroxylation of the substrate phenylalanine (black) to the product tyrosine (black), BH_4 leaves the enzyme as a 4a-hydroxy intermediate, BH_4-4a-carbinolamine, and is regenerated back to BH_4 by two external enzymes, PCD/DCoH1 and DHPR. In the absence of sufficient DHPR activity, the quinoid 6,7[8H] dihydrobiopterin intermediate rearranges non-enzymatically to 7,8-dihydrobiopterin (broken line) which is no longer a substrate of DHPR, but requires DHFR to be reduced back to BH_4.

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also occurs non-enzymatically, but at a rate that is, at least in the liver, insufficient to maintain BH_4 in the reduced state [40]. As a consequence, liver PCD/DCoH1 deficiency in humans causes BH_4-4a-carbinolamine to be rearranged via a spiro structure intermediate to dihydroprimapterin (or 7-substituted dihydrobiopterin) that is excreted in the urine [41].

The final conversion of quinoid dihydrobiopterin into BH_4 is carried out by the dimeric DHPR in a NADH-dependent reaction (Figure 4). Although the crystallographic structure of the DHPR–NADH binary complex was solved, the location of the active sites is not known from these studies. This final reaction of the regeneration pathway involves direct hydride transfer from the reduced nicotinamide ring to the quinoid dihydrobiopterin by DHPR. The reaction is supported by the proposed enzyme mechanism of NAD(P)H-dependent reductases and by the lack of detectable prosthetic groups such as flavin or metal ions [42].

BH_4 homoeostasis

Regulation of BH_4 by the biosynthetic enzyme GTPCH and its complex with GFRP is best studied in endothelial tissue or endothelial-derived cells. Little is known about modulation of GTPCH activity in other organs such as liver or brain. Moreover, it is unclear how homoeostasis and bioavailability of the BH_4 cofactor is regulated, which involves biosynthesis, distribution, transport (uptake and release), interchange of oxidation state (BH_4 or 7,8-dihydrobiopterin) and regeneration of biopterin etc. Yet, the bioavailability and/or oxidation state of biopterin appears to influence, directly or indirectly, various physiological processes, including vascular function/disfunction, hypertension, diabetes, hypercholesterolaemia, as well as several metabolic-biosynthetic pathways such as those for blood phenylalanine, monoamine neurotransmitters, folates and homocysteine. Some of these processes are influenced greatly by cofactor bioavailability, oxidative stress and NOS decoupling (see below), but not all of them, such as transport and effect of BH_4 on aromatic amino acid hydroxylases. In general, BH_4 is labile in solution at physiological pH and can readily react with O_2 to produce free radicals, thus generating oxidative species in cells, including superoxide, H_2O_2 and peroxynitrite [43]. On the other hand, BH_4 can also protect cells against oxidative damage by scavenging radicals [44]. Spontaneous oxygen reactivity, a chaperone-like effect on protein synthesis, superoxide formation by NOS uncoupling and other functions of BH_4 all hint towards the importance of the concentration of BH_4 as a critical factor that determines the beneficial or adverse effects *in vivo*. In this context, toxicology of BH_4 is also an issue to consider when experimental treatment studies with cells or animals are conducted (for an overview, see Blau and Erlandsen [45]).

BH_4-DEPENDENT REACTIONS

Mechanisms of BH_4-dependent enzymes

All known BH_4-dependent reactions are mixed-function mono-oxygenases, i.e. the oxygen molecule is used as a co-substrate, one oxygen atom is incorporated into the substrate and the second oxygen atom ends up in a water molecule (Figure 1). In close analogy to the protein sequence homology (Figure 1), three classes of BH_4-dependent enzymatic reactions can be distinguished. First, the aromatic amino acid hydroxylases contain a catalytically active non-haem iron. BH_4 is involved in oxygen activation, donates two electrons and leaves the enzymatic reaction as a 4a-hydroxy derivative and is regenerated back to BH_4 by two external enzymes. Secondly, the NOSs contain a catalytically important haem iron. BH_4 is not involved in oxygen activation, donates only one electron for each of the two-step reactions from arginine to citrulline, and is regenerated on NOSs without the need for external enzymes. Thirdly, AGMO is assumed from sequence homology to contain a di-iron centre, but otherwise shares many features of the mechanism of aromatic amino acid hydroxylases.

Aromatic amino acid hydroxylases

PAH was the first enzyme recognized to depend on BH_4 ([46], reviewed by Kaufman [47]). On the basis of detailed biochemical (reviewed by Fitzpatrick [48]) and structural data [49], the stimulation of PAH by BH_4 is generally accepted to proceed as displayed in a simplified way in Figure 4. A crucial feature in this mechanism is a non-haem iron in the active site, which is bound by two histidine residues (His^{285} and His^{290}, numbers refer to PAH) and Glu^{330}. BH_4 and phenylalanine bind in close vicinity to this non-haem iron. The conserved Glu^{286} has been shown to be important for the BH_4 interaction [50,51]. The conserved His^{285} and His^{290} as well as the conserved Glu^{286} form the core of the Prosite protein motif recognizing all BH_4-dependent aromatic amino acid hydroxylases {P-D-X_2-H-[DE]-[LIVF]-[LIVMFY]-G-H-[LIVMC]-[PA], where X is any amino acid; PDOC00316} [52]. Molecular oxygen binds to the non-haem iron and is activated with the help of BH_4, which forms a bond to the oxygen via the 4a position (Figure 4). The O–O bond is then cleaved, and a highly activated iron–oxo species, which is frequently designated as an Fe(IV)–oxo complex, hydroxylates phenylalanine. BH_4 leaves the reaction and the enzyme as its 4a-hydroxy derivative, and is then regenerated by the action of two external enzymes as discussed above. PCD/DCoH1 facilitates the elimination of water from the 4a-hydroxy-derivative to form the quinoid 6,7-[8H]-dihydrobiopterin, which is finally converted back into BH_4 by DHPR (Figure 4). In the absence of this regeneration, the hydroxylation reaction is stoichiometric for BH_4, i.e. one BH_4 added yields one tyrosine residue formed.

In the absence of sufficient DHPR activity, the quinoid 6,7[8H]dihydrobiopterin rearranges non-enzymatically within minutes to 7,8-dihydrobiopterin. This is no longer a substrate of DHPR, but may be reduced to BH_4 by DHFR (Figure 4). This additional regeneration, however, cannot compensate for the loss of DHPR activity, as is clearly shown in a malignant clinical picture of DHPR deficiency [53].

Other aromatic amino acid hydroxylases, such as TH and the two TPHs TPH1 and TPH2, share many features with PAH with respect to the reaction mechanism [54,55], to BH_4 [51] and substrate [56] binding.

NOSs

It has been more than 30 years since the discovery of pteridine dependence for PAH [46] that a newly discovered enzymatic reaction, that of NOS, was found to critically depend on BH_4 [57,59], which was found to be required for both steps of this complex reaction [60,61]. The first sequence of a NOS [62] then revealed the lack of any protein homology with aromatic amino acid hydroxylases. Biochemically, NOSs behave very differently to aromatic amino acid hydroxylases. Added BH_4 was able to catalyse more than 15 [59,63] to 26 molecules of product rather than one as with aromatic amino acid hydroxylases in the absence of regeneration enzymes, and no external regeneration enzymes were required. Another striking feature was the high affinity of

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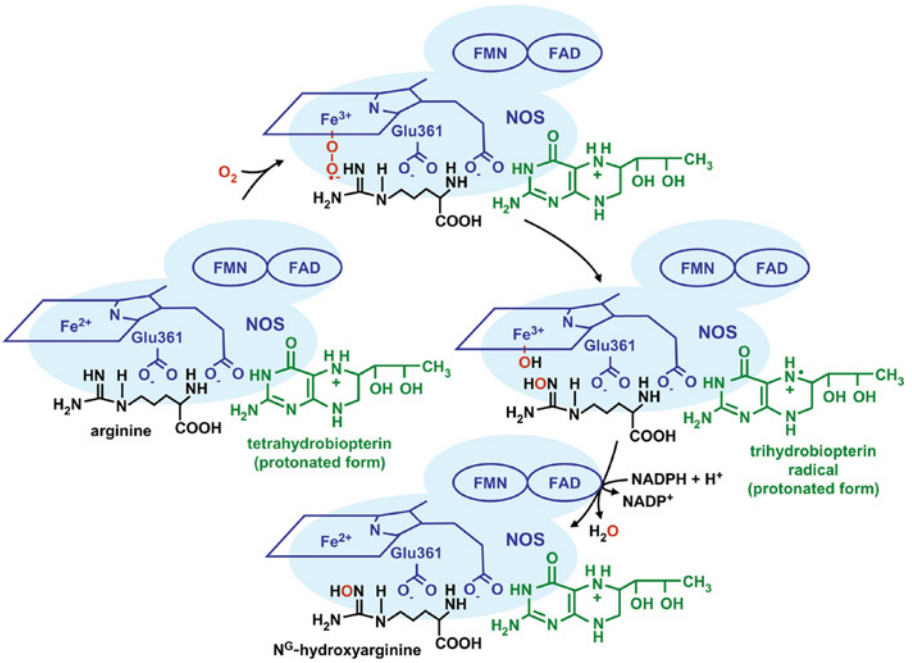

Figure 5 Simplified scheme of the NOS reaction

Only the first half of the NOS reaction, the conversion of arginine into N^G-hydroxyarginine (both shown in black) is shown. BH_4 (green) is positioned distant to the haem iron which activates the oxygen (red), but still in a position close enough to provide electrons to the haem via the porphyrin backbone (blue). BH_4 plays no role in activation of the oxygen, but supplies one electron to the reaction, yielding a trihydrobiopterin radical cation. This is regenerated to BH_4 when bound to the enzyme by an electron provided from NADPH via FAD and FMN of the reductase domain. BH_4 presumably also supplies a proton to the active site via a proton bridge including the haem proprionyl carboxylate (blue) and a glutamyl carboxylate (blue) side chain of NOS. N^G-hydroxyarginine also remains bound to the enzyme, and the catalytic cycle is completed by a second mono-oxygenase reaction of a similar mechanism, again with the participation of BH_4 to yield citrulline and NO (not shown, see also Figure 1).

BH_4 to NOSs, and the higher selectivity of this enzyme class to the stereochemistry of the side chain at position 6 of the pteridine ring [57,58]. BH_4 was found to stabilize the dimeric form of NOS [64,65]. Biochemical evidence argued against a direct role for BH_4 in oxygen activation of the enzyme [66], but argued for a redox-active contribution to catalysis by reductive activation of the haem–oxygen complex [67]. A crucial step in the elucidation of the reaction mechanism was the detection of a trihydrobiopterin radical [68–70]. Together with structural data on the oxygenase domain of the NOSs [71–73], a consistent picture of BH_4 simulation of NOS emerged (reviewed in [74]), which is shown in Figure 5 in a simplified form.

As in aromatic amino acid hydroxylases, iron is crucial for catalysis. In NOSs, however, the iron is present as a haem iron. BH_4 binds distant to the iron atom, but still close enough to the porphyrin ring to allow interaction with the haem (Figure 5). Haem and BH_4 are found in the oxygenase domain of the enzyme, the additional cofactors FAD and FMN bind to the reductase domain, which donates electrons from NADPH to the oxygenase domain. NOSs are active as dimers, and the reductase domain of one dimer interacts with the oxygenase domain of the other dimer [75] (see the review by Stuehr [76] for details). For simplicity, this detail is omitted in Figure 5. BH_4 is bound to NOSs as a cation [72]. Oxygen binds to the haem iron, but, in contrast with aromatic amino acid hydroxylases, BH_4 is not involved in activation of the oxygen. It does, however, donate an electron and a proton to the haem iron–oxygen complex during catalysis to form a protonated trihydrobiopterin radical [70]. The proton transfer is enabled by a proton bridge involving N-3 of the pterin, the haem proprionate and a conserved glutamate residue of the oxygenase domain [74]. The protonated BH_4 cofactor and the Fe^{2+} haem are then regenerated by electrons supplied from NADPH via the reductase domain and the FAD–FMN couple. After this first step of the reaction yielding N^G-hydroxyarginine, which remains bound to the enzyme, citrulline and NO are formed from N^G-hydroxyarginine by a second similar round of catalysis, again with the participation of BH_4.

All three isoforms share this reaction mechanism with respect to BH_4, and have a considerable sequence homology (Figure 1). The most prominent differences in the sequences among the three isoforms of NOS is a spacer sequence between the oxygenase and the reductase domains, which is contained in NOS1 and NOS3, but is lacking in NOS2 sequences. This spacer sequence is related to the dependence of NOS1 and NOS3 activity on the concentration of free calcium ions [77]. NOS2, which lacks the spacer sequence, does not need a calcium trigger for producing NO [76].

AGMO

Only a few years after the first description of the pteridine dependence of PAH, a tetrahydropteridine-requiring enzyme system for the oxidation of glyceryl ethers was described [78]. Knowledge on this enzyme remains limited because, so far, it has resisted all protein purification attempts. Since it is an integral membrane enzyme, it needs to be solubilized before purification. This process destabilizes this enzyme so that it rapidly loses activity. The recent assignment of its sequence to TMEM195, a predicted membrane protein with thus far unknown function, required bioinformatic selection of candidates, recombinant expression in cells and monitoring of the enzymatic activity [79]. The suggested standard name for this enzyme is AGMO, but it is also known as glyceryl ether mono-oxygenase or ether lipid oxidase.

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With respect to the biochemistry of the stimulation of its activity, AGMO shares many features with aromatic amino acid hydroxylases. BH_4 is a stoichiometric reactant of the reaction, and thus requires regeneration by external enzymes [80]. Similar to aromatic amino acid hydroxylases, treatment of homogenates containing the enzyme activity with the iron chelator 1,10-phenanthroline inhibit the activity, compatible with the presence of a catalytically important non-haem iron [81]. Selectivity for the side chain at position 6 of the pteridine ring [82] and biochemical behaviour towards BH_4 analogues [81] was also similar for AGMO and aromatic amino acid hydroxylases, but different from NOSs. Consistent with these findings, lack of inhibition of the enzyme by CO also argued against a cytochrome P450-like haem-containing enzyme [83].

The AGMO sequence contains no characteristic haem-binding site but instead has a fatty acid hydroxylase motif, which is also found in fatty acid hydroxylases and desaturases. This motif comprises eight conserved histidine residues [84], and each of these is required for enzymatic activity in AGMO [79]. Fatty acid hydroxylase motif-containing enzymes are thought to use a di-iron centre for catalysis [85], and none of the membrane enzymes of this class has ever been purified to homogeneity, presumable due to the instability upon solubilization. Except for AGMO, no BH_4 dependence of a fatty acid hydroxylase type of enzyme has been described. On the other hand, all di-iron hydroxylases characterized include a multi-subunit hydroxylase, electron-transfer proteins and a cofactor-less effector protein that is unique to the di-iron hydroxylase family [86]. Future work will have to show whether BH_4 can substitute one or more of these additional components in AGMO.

Effect on other enzymes and additional roles

One of the earliest discovered cellular functions of BH_4 was the growth factor of *Crithidia fasciculata*, and this was initially used to measure biopterins in various body fluids and tissues. Further observations suggested a proliferative activity of BH_4 in haemopoietic cells [87,88]. Exogenous BH_4 was found to stimulate DNA synthesis and induce proliferation of some mouse erythroleukaemia clonal cell lines. BH_4 and sepiapterin also enhanced the proliferation of SV40 (simian virus 40)-transformed human fibroblasts, rat C6 glioma cells and PC12 cells. Subsequently, it has been shown that EGF (epidermal growth factor), NGF (nerve growth factor) and IGF-1 (insulin-like growth factor-1) increased the proliferation of rat PC12 cells through obligatory elevation of intracellular BH_4 [89,90].

Besides its proliferative activity, BH_4 has been suggested to act as a self-protecting factor for NO toxicity with generation of superoxide in NO-producing neurons [91]. Indeed, strong scavenging activity of BH_4 against superoxide anion radicals was shown in both xanthine oxidase and rat macrophages/PMA radical-generating systems, and the authors suggested that BH_4 might be useful in the treatment of various diseases where pathogenesis is actively oxygen-related [92]. In another series of experiments, Shimizu et al. [93] demonstrated that *S*-nitroso-*N*-acetyl-DL-penicillamine (an NO donor)-induced endothelial cell death can be prevented by increasing cellular levels of BH_4. This finding was an early observation which suggested the cytotoxicity of NO involving H_2O_2 production, and that scavenging of H_2O_2 by BH_4 may be at least one of the mechanisms by which BH_4 reduces NO-induced endothelial cell death. In contrast, Cho et al. [94] hypothesized that ischaemia increases the intracellular BH_4 levels and that the increased BH_4 level plays a critical role in selective neuronal injury through NOS activation. Using a selective inhibitor of GTPCH in animals exposed to transient forebrain ischaemia, they demonstrated a marked reduction of BH_4 levels, NADPH-diaphorase activity and caspase-3 gene expression in the CA1 hippocampus. Moreover, delayed neuronal injury in the CA1 hippocampal region was significantly attenuated by the GTPCH inhibitor. These data, in contrast with those by Shimizu et al. [93], suggested that a blockade of BH_4 biosynthesis may provide novel strategies for neuroprotection.

Modulation of BH_4-dependent gene expression was put forward based on several observations. In the GTPCH-deficient (*hph-1*) mouse, up-regulation of liver *Pah* and brain *Th* gene expression upon BH_4 treatment was reported; however, no such change was found in the wild-type control mice [95]. In smooth muscle cells, post-transcriptional stabilization of *NOS2* mRNA was observed indirectly upon varying endogenous BH_4 levels [96]. An effect of BH_4 supplementation on *PAH* gene expression was also discussed in BH_4-responsive phenylketonuria (see below) [97]; however, gene expression or mRNA stability for PAH was not changed, at least not in transgenic mice with different cofactor concentrations present in the liver [98].

In addition to its other functions, BH_4 enhances the release of dopamine and serotonin in the rat striatum when administered locally through the dialysis membrane [99,100]. The enhancement of dopamine release persisted even when dopamine biosynthesis or dopamine re-uptake was completely blocked, but it was abolished when blockers of voltage-dependent Na^+ or Ca^{2+} channels were administered with BH_4. Further experiments using selective inhibitors of TH and NOS demonstrated that BH_4 stimulates dopamine release directly, independent of its cofactor action on TH and NOS, by acting from the outside of neurons [101]. The exact mechanism is not entirely clear, but it has been shown that arginine also induces a concentration-dependent increase of dopamine release in rat striatum slices, and that it is dependent on the presence of BH_4.

Another role for BH_4 on PAH and TH (but not on TPHs) is the chemical chaperone effect preventing protein misfolding and inactivation during dimer/tetramer expression, and protection from proteolytic cleavage [98,102,103]. Thus BH_4 plays a central regulatory role in the phenylalanine hydroxylating system. The only other known BH_4-requiring enzymes in liver, AGMO and NOS, are present in relatively low amounts, and PAH (subunit) and BH_4 concentrations in liver are approximately equal (PAH, $\sim 9\ \mu$mol/l; and BH_4, 5–10 μmol/l [98,102,104–108]). Nevertheless, considering the K_m for the cofactor in the PAH reaction, which was estimated to be 25–30 μmol/l [106,108], BH_4 is limiting and subsaturating at least in the liver [98,109] (with consequences for BH_4-responsive phenylketonuria; see below). There is no evidence that TH and TPHs are regulated by substrate-activated mechanisms similar to those that regulate PAH. All aromatic amino acid hydroxylases are inhibited by catecholamines, but only the inhibition of human TH is competitive with respect to the BH_4 cofactor, and it has been shown that the cofactor can directly displace dopamine from the enzyme active site [110].

PATHOLOGY OF BH_4

Inherited disorders in BH_4 metabolism

BH_4 deficiencies, a group of rare inherited neurological diseases with catecholamine and serotonin deficiency, may present phenotypically with or without hyperphenylalaninaemia [111]. It is a heterogenous group of diseases, affecting either all organs, including the central nervous system, or only the peripheral hepatic PAH system. BH_4 deficiency presenting with

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Table 2 Inherited disorders of BH_4 metabolism and biochemical markers for differential diagnosis

↑, elevated; ↓, lowered; 5HIAA, 5-hydroxyindoleacetic acid; Bio, biopterin; HVA, homovanillic acid; n, normal; Neo, neopterin; OMIM, Online Mendelian Inheritance in Man; Phe, phenylalanine; Pri, primapterin (7-biopterin); Sep, sepiapterin.

		Blood	Urine			Cerebrospinal fluid					
Disease	OMIM number	Phe	Neo	Bio	Pri	Neo	Bio	Sep	5HIAA	HVA	Folates
GTPCH deficiency (autosomal recessive)	233910	↑	↓	↓	n	↓	↓	n	↓	↓	n
PTPS deficiency	261640	↑	↑	↓	n	↑	↓	n	↓	↓	n
PCD deficiency	264070	↑	↑	n-↓	↑	n	n	n	n	n	n
DHPR deficiency	261630	↑	n	n-↑	n	n	↑*	n	↓	↓	↓
GTPCH deficiency (autosomal dominant; DRD)	128230	n	n	n	n	↓	↓	n	n-↓	↓	n
SR deficiency	612716	n	n	n	n	n	↑*	↑	↓	↓	n

*7,8-dihydrobiopterin

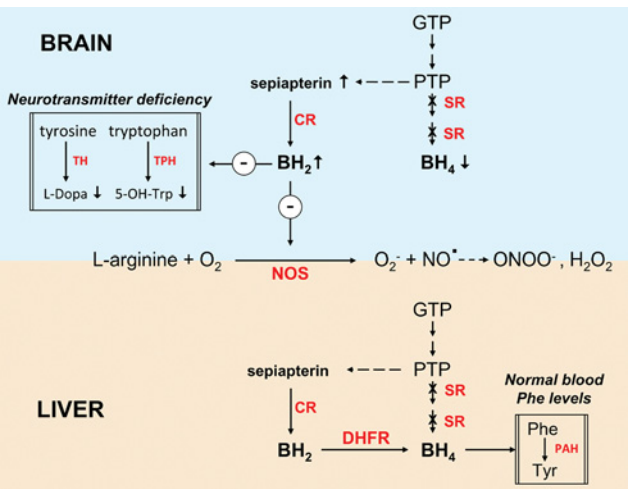

Figure 6 Consequences of SR deficiency on neuronal NOS uncoupling

In the absence of SR activity and due to a low DHFR activity in the brain, sepiapterin and 7,8-dihydrobiopterin (BH_2) accumulate and uncouple NOS. Uncoupling of NOS is potentiated by low BH_4 concentrations and results in superoxide and peroxinitrite production, both potentially neurotoxic. In addition, low BH_4 concentrations and high 7,8-dihydrobiopterin inhibit production of catecholamines and serotonin in the central nervous system. In peripheral tissues such as liver, BH_4 biosynthesis is assured by the high DHFR activity and hepatic PAH functions normally. PTP, 6-pyruvoyltetrahydropterin.

hyperphenylalaninaemia can be caused by mutations in genes encoding the enzymes involved in its biosynthesis (GTPCH and PTPS) or regeneration (PCD/DCoH1 and DHPR) [53]. The mutations are all inherited in an autosomal recessive fashion. Biochemical, clinical and molecular data of patients with BH_4 deficiencies are tabulated in the BIODEF and BIOMDB databases and are available on the Internet (http://www.biopku.org). Depending on the enzyme defect and the mode of inheritance, patients are diagnosed by different analytical and biochemical approaches. A few simple tests (pterins in blood, DHPR activity in blood and BH_4 loading tests) discriminate between classical phenylketonuria and cofactor defects and additional investigations on neurotransmitter metabolites in cerebrospinal fluid serve to define the disease (Table 2).

The two forms of BH_4 deficiency that may occur without hyperphenylalaninaemia are an autosomal dominantly inherited form of GTPCH deficiency [or DRD (dopa-responsive dystonia)], initially described as Segawa disease [112], and SR deficiency [113]. In SR-deficient patients, alternative reductases (e.g. aldose-ketose reductases and 3α-hydroxysteroid dehydrogenase 2; see also Figures 3 and 6) substitute the function of SR in hepatic tissue.

Patients presenting with hyperphenylalaninaemia are usually detected through the neonatal screening programmes for phenylketonuria, whereas those presenting without hyperphenylalaninaemia are recognized either by the typical clinical signs and symptoms or by analysis of neurotransmitter metabolites and pterins in cerebrospinal fluid and by investigation of cultured skin fibroblasts. The clinical course is similar in untreated patients with severe forms of autosomal recessive GTPCH, PTPS and DHPR deficiencies. The variable but common symptoms include mental retardation, convulsions (grand mal or myoclonic attacks), disturbance of tone and posture, drowsiness, irritability, abnormal movements, recurrent hyperthermia without infections, hypersalivation and swallowing difficulties [114]. The absence of clinical signs, theoretically, defines phenotypically mild peripheral forms. In contrast with severe BH_4 deficiency (treatment with L-dopa/carbidopa and 5-hydroxytryptophan plus BH_4), these patients can be treated with BH_4 alone.

In autosomal dominant GTPCH deficiency, dystonic posture or movement of one limb with diurnal fluctuation is typically present and may exacerbate with age. There is a marked and sustained response to low doses of L-dopa/carbidopa without any side effects. The clinical features of patients with SR deficiency are similar to those in autosomal dominant GTPCH deficiency; however, additional neurological hallmarks, including cerebral palsy, are frequently noted. We demonstrated that cytokine-stimulated fibroblasts are useful as a system to measure neopterin and biopterin production and GTPCH activity [115]. After stimulation for 24 h with interferon-γ and tumour necrosis factor-α, the concentrations of neopterin and biopterin were extremely low compared with control fibroblasts.

BH_4 treatment of vascular dysfunction and ischaemia-reperfusion injury

When studying the products of purified NOS, formation of H_2O_2 was detected at suboptimal concentrations of BH_4 or L-arginine [59]. Depletion of BH_4 in canine arteries also led to the formation of H_2O_2 in arterial rings [116]. Quickly, it was realized that conditions harming endothelium-dependent relaxation such as smoking [117], ischaemia/reperfusion injury [118] or diabetes [119] might be caused by lowered concentrations of BH_4 and, therefore, might be improved by treatment with BH_4 or with sepiapterin, which is converted into BH_4 by the consecutive action of SR and DHFR. A simplified version of the current view on the action of BH_4 in endothelial dysfunction is shown in Figure 7. For further details, see previous reviews [19,120–123].

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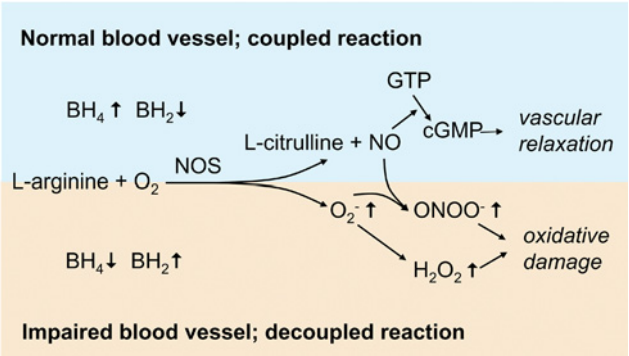

Figure 7 Simplified scheme of the induction of vascular damage by altered BH_4 to 7,8-dihydrobiopterin ratios

A simplified form of a widely used explanation of the relationship of BH_4 depletion in blood vessels by alteration of the BH_4 to 7,8-dihydrobiopterin (BH_2) ratio caused by smoking, diabetes or ischaemia/reperfusion injury is shown. In normal vessels, hormonal signals such as acetylcholine lead to a transient increase of free Ca^{2+} ions, and this stimulates the formation of NO from NOS which in turn stimulates guanylate cyclase to form cGMP from GTP. The cGMP signal then causes the vessel to relax. In the impaired blood vessel, in contrast, NOS forms reactive oxygen species due to the insufficient supply with BH_4 in relation to 7,8-dihydrobiopterin. The treatment with BH_4 corrects this insufficiency and therefore restores formation of NO and diminishes reactive oxygen species formation. Although most studies assume that endothelial NOS is the target of the BH_4 treatment effect on vascular dysfunction, it remains to be firmly proven whether this holds true.

Is endothelial NOS the target of BH_4 therapy for improvement of impaired vascular function?

Although it is generally assumed that endothelial NOS is the target of BH_4 action in vascular dysfunction, firm proof of this assumption, such as a lack of a BH_4 effect in endothelial NOS-deficient mice, is still missing. Potentially, BH_4 might mediate its action by any of the reactions depending on it as a cofactor (Figure 1), or even more generally by acting as an antioxidant. One possibility to address this point is the comparison of various tetrahydropteridine derivatives. As previously mentioned, NOSs have a higher selectivity for the stereochemistry of the side chain at C-6 of the tetrahydropteridine ring as compared with other BH_4-dependent enzymes. Therefore comparison of the action of BH_4 with tetrahydroneopterin, which stimulates all BH_4-dependent enzymes, except NOSs, can discriminate NOSs from other actions, provided it has similar pharmacokinetics as BH_4, which have yet to be demonstrated. In line with the assumption of NOS as a drug target, tetrahydroneopterin could neither substitute BH_4 for the correction of vascular dysfunction in smokers [124], nor for diminishing superoxide production and correcting of blood pressure in caveolin-1-deficient mice [125] as well as in a mouse model of hypertension [126]. These data suggest NOS as the target of BH_4 treatment. One study in human forearms, however, found a similar action of different tetrahydropteridine derivatives, suggesting an antioxidant role as the mechanism [127]. Although it is tempting to assume that the endothelial isoform of NOS is the target, some reports suggest an important role for neuronal NOS in the vasculature [128]. Neuronal NOS as a target would also be compatible with the experimental observations. In rat hepatic ischaemia/reperfusion injury, the inducible isoform was suggested as a target of BH_4 action [129]. Finally, BH_4 treatment [130] or myocardial GTPCH overexpression [131] resulted in prolonged cardiac allograft survival which is difficult to explain as a result of action on endothelial NOS only.

Altered BH_4 concentrations are associated with endothelial dysfunction

Depletion of BH_4 by inhibition of its biosynthesis caused the formation of H_2O_2 in aortic rings [116] and impaired vascular function of rat hearts [132]. In a collection of patients entering hospital for coronary angiography because of atypical chest pain, erythrocyte DHPR activity and plasma BH_4 to (7,8-dihydrobiopterin + biopterin) ratio was significantly lower in patients presenting with insulin resistance [133]. In 168 patients with coronary artery disease, the plasma BH_4 to 7,8-dihydrobiopterin ratio was strongly correlated with endothelial dysfunction [134]. In a proteomic comparison of spontaneously hypertensive rats and normal rats, DHPR was found to be lowered in the hypertensive animals, and this was accompanied by lowered BH_4 levels [135]. Rat strains that are more resistant to myocyardial ischaemia have a higher BH_4 to 7,8-dihydrobiopterin ratio [136]. Ischaemic preconditioning of the main coronary artery in rats led to an up-regulation of BH_4 synthesis and resulted in acquisition of ischaemic tolerance [137]. In aortae from partially BH_4-deficient *hph-1* mice, but not from control mice, catalase inhibited the relaxation of aortae, indicating the formation of H_2O_2 upon stimulation with acetylcholine. This difference disappeared upon BH_4 treatment [138].

Induction of hypertension or ischaemia–reperfusion injury results in BH_4 depletion

Induction of hypertension in mice by deoxycorticosterone acetate salt led to a decrease in BH_4 and an increase in 7,8-dihydrobiopterin + biopterin [126,139]. Since this change was largely absent in mice missing a functional NADPH oxidase, oxidation of biopterin by reactive oxygen species is a probable mechanism leading to this change [139]. This observation is also compatible with previous studies comparing arteries from diabetics with non-diabetic patients [140]. 26S-proteasome-mediated degradation of GTPCH has been suggested as additional mechanism depleting BH_4 in diabetes [141]. Once BH_4 availability is suboptimal, reactive oxygen species are formed by NOS, and this is thought to amplify BH_4 degradation. Superoxide dismutase, but not catalase, prevented BH_4 degradation in mice rendered hypertensive with angiotensin II [142]. Ischaemia/reperfusion injury diminished the BH_4 to (7,8-dihydrobiopterin + biopterin) ratio in mouse pancreas [143] and rat heart [144], and decreased BH_4 levels in rat kidney [145].

Treatment of humans with BH_4 to correct vascular dysfunction

More than 288 reports have been published to date on studies trying to ameliorate vascular dysfunction by treatment with BH_4 in animals and in humans. Most studies in humans were carried out by infusion of solutions to, e.g., the forearm [124,146–156]. Two oral treatment studies to correct vascular dysfunction in humans with hypercholesterolaemia or poorly controlled hypertension were carried out with success using daily doses of approximately 5–10 mg of BH_4/kg [157–158]. Only a few human studies failed to observe positive effects of BH_4 treatment on endothelial function [159,160].

Endothelium-specific overexpression of GTPCH mimics the effects of BH_4 treatment of vascular dysfunction

Targeted overexpression of GTPCH in the endothelium elegantly demonstrated the role of BH_4 availability for the development of atherosclerosis in *ApoE* (apolipoprotein E)-ko (knockout) mice

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[161], for maintaining a physiological platelet state in diabetic mice [162], for endothelial function in diabetic mice [163] and for lowering blood pressure in salt-sensitive low-renin hypertension [164]. In addition, GTPCH overexpression accelerated wound healing in Type 1 diabetic mice by enhancing calcium-dependent NOS activity and suppressing oxidative stress [165].

Does ascorbate improve vascular function by stabilizing BH_4?

Ascorbate blunted the improvement of vascular dysfunction by BH_4 in human smokers [124]. Ascorbate has been shown to stabilize BH_4 levels in cultured endothelial cells [166,167] and to increase BH_4 plasma concentrations in humans [168]. The observed treatment effects of ascorbate in hypertension [169] might therefore be related to BH_4 concentrations in the vascular wall. In *ApoE*-deficient mice, ascorbate protected endothelial function [170]. In *ApoE*-deficient mice with transgenic endothelial overexpression of endothelial NOS, however, ascorbate failed to reverse atherosclerotic lesion formation. Targeted overexpression of GTPCH in the endothelium was required to achieve this goal [171]. In ischaemic rat hearts, BH_4, but not ascorbate, was able to correct impaired NO synthesis [172].

BH_4 treatment of phenylketonuria

Phenylketonuria is a genetic disorder characterized by a deficiency of the hepatic enzyme PAH, causing elevated concentrations of phenylalanine in the blood and brain [173]. Higher blood phenylalanine concentrations are associated with more severe disease and a greater risk of neurological impairment and, as such, require more urgent treatment. Until recently, the main treatment option in phenylketonuria involved putting patients on a restricted low-phenylalanine diet. Today, a new treatment, BH_4 (sapropterin dihydrochloride), is available, which may provide good phenylalanine control in the patients who respond to oral administration of BH_4, with the possibility of making adjustments towards a more normal diet [174]. Although phenylketonuria is characterized by a defect in the PAH enzyme, residual enzymatic activity may be present in some patients. Thus BH_4 may act like a chemical chaperone to promote the normal metabolism of phenylalanine and lower its concentration in the blood in a subset of patients who are BH_4-responsive [102]. A specific BH_4-responsive genotype is characteristic for these patients [175].

ANIMAL MODELS TO STUDY DISEASE PATHOLOGY AND BH_4 DEFICIENCIES

Mouse models have become an essential tool in biomedical research for studying gene function and modelling human genetic disorders, including BH_4 deficiency and its pathology. Mouse models are available for most of the primary defects in BH_4 metabolism, such as cofactor biosynthesis and regeneration, but also for BH_4 cofactor-dependent enzymes (see Table 1). Regarding primary defects, the *hph-1* mouse and different ko/ki (knockin) and transgenic strains for PTPS exist, as well as targeted deletions for the murine genes encoding SR, PCD/DCoH1 and DHPR. Despite attempts from several research groups, the generation of a mouse model for the recessive form of GTPCH deficiency by homologous recombination of the murine *Gch1* gene in embryonic stem cells was not successful (B. Thöny, unpublished work). As an alternative, siRNA (small interfering RNA)-mediated knock-down of the (bovine) *Gch1* gene was reported under cell culture conditions [176]. The *hph-1* mouse was produced by screening ethylnitrosourea-treated mice for the presence of hyperphenylalaninaemia, and subsequent biochemical analysis revealed that this strain provides an animal model for DRD [177,178]. Unfortunately, no DNA mutation could be identified thus far, although genomic mapping narrowed the site of mutation to an interval containing the mouse *Gch1* gene [179]. This chemically induced *hph-1* mutant strain has a mild, neonatal and transient form of hyperphenylalaninaemia with lowered GTPCH activity and BH_4 cofactor levels in the liver. In the brain, the mouse showed low levels of BH_4, catecholamines, serotonin and TH activity. Further 'refinement' for the DRD model was performed by transgenic introduction into the *Pts* homozygous ko background ($Pts^{-/-}$; see below) of the *PTS*-cDNA under the control of the dopamine β-hydroxylase promoter. The resulting mice exhibited hyperphenylalaninaemia and limited expression of PTPS in dopaminergic neurons, but not in noradrenergic neurons. It was concluded that the biochemical and pathological changes in these mice were similar to those in human BH_4 deficiency and DRD, resulting in mice with severely reduced striatal BH_4 and TH, but normal noradrenaline and adrenaline production [180,181].

A conditional ko by floxing the murine *Gchfr* gene is in preparation to study GFRP function (D. Adamsen, T. Scherer, N. Blau and B. Thöny, unpublished work).

When the first mouse unable to synthesize BH_4 was generated by completely knocking out the *Pts* gene, it was found that $Pts^{-/-}$ newborns died immediately after birth, which for human patients with PTPS deficiency is not so dramatic, although PTPS deficiency in its severest form is eventually lethal [182,183]. The monoamine neurotransmitters are required during embryogenesis, but mice die only after birth as the BH_4 cofactor is supplied by the mother during pregnancy. Repeated administration of BH_4 in combination with the neurotransmitter precursors L-dopa/carbidopa and 5-hydroxytryptophan rescued the perinatal lethality with 100 % survival of $Pts^{-/-}$ mice, but resulted in dwarfism [183]. A detailed analysis of these dwarf mice revealed that BH_4 levels normalized in brain and other tissues, and that hepatic PAH and neuronal TPH and NOS had normal activity. However, brain TH immunoreactivity, enzyme activity and dopamine levels remained very low, despite normal TH gene expression [182]. In addition, the pituitary-derived growth hormone and the THS-dependent thyroxine T4 were both normal, indicating that the pituitary gland is normally developed, whereas IGF-1 levels were only 15 % of wild-type controls. From these results, in combination with the observations discussed above, it was hypothesized that the dwarfism was due to limited appetite resulting in chronic under-nutrition, with consequent low IGF-1 because of the selectively low TH and dopamine levels [183]. In parallel, abnormal reduction of plasma IGF-1 has been observed in newborn patients with BH_4 deficiency due to PTPS mutations. As lack of BH_4 leads to brain TH but not TPH protein depletion, it was concluded that catecholamine and serotonin synthesis are differently regulated by BH_4 [182]. Currently, more mouse models for BH_4 deficiency by targeting the *Pts* gene are under investigation: a *Pts*-R15C-ki (*Pts*-ki) mouse, equivalent to the human *PTS*-R16C with a mild peripheral phenotype, and compound heterozygous *Pts*-ki/ko mice by breeding the ki allele into *Pts*-ko (null) mutant. Homozygous *Pts*-ki mice showed no hyperphenylalaninaemia or monoamine neurotransmitter deficiency, but had reduced PTPS activity and elevated neopterin, and abnormal fat distribution. In comparison, newborn *Pts*-ki/ko exhibited reduced BH_4 and PTPS activity in liver and brain, elevated neopterin, monoamine neurotransmitter

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deficiency in brain, and mild hyperphenylalaninaemia (~0.9 mmol/l) [184]. Although still under investigation, these new *Pts* transgenic mouse groups might represent, to some extent, models for the heterogeneous group of human BH_4 deficiencies due to mutant PTPS where approximately a quarter of patients with PTPS deficiency have a mild or peripheral phenotype with isolated hyperphenylalaninaemia and lowered hepatic BH_4, and the rest have a severe or central phenotype with additional depletion of BH_4, catecholamine and serotonin neurotransmitters in the central nervous system, besides systemically elevated neopterin as a precursor of BH_4 [53].

Animal models for SR, PCD/DCoH1 and DHPR deficiencies were all generated by homologous recombination in embryonic stem cells, but these mice only appeared to partially represent the corresponding human counterparts. SR deficiency, an autosomal recessive disease featuring BH_4-dependent neurotransmitter deficiency with impaired body movements but without hyperphenylalaninaemia, presents in the mouse with a complete *Spr*-gene ko with phenylketonuria, which is a distinct contrast from human patients [113,185,186]. Mice lacking PCD/DCoH1 are viable and fertile but display hyperphenylalaninaemia and a predisposition to cataract formation. Owing to the presence of the mouse DCoH2 homologue, partial complementation of PCD/DCoH1 activity cannot be excluded [187,188]. A DHPR (*Qdpr*)-ko mouse was produced with complete absence of DHPR activity. This mouse mutant was viable and had normal BH_4 levels in liver in addition to an abnormally elevated blood phenylalanine concentration and low brain neurotransmitter levels [189].

Besides animal models for BH_4 deficiency, an endothelium-specific GTPCH overexpressing mouse was generated with elevated total biopterin content in lung, heart and aorta, but normal levels in liver and plasma [190]. This transgenic mouse model turned out to be a valuable tool in studying endothelial NOS function and vascular disease states [191,192].

Mouse models for BH_4 cofactor-dependent enzyme deficiencies have been reviewed elsewhere and will not be described in detail in the present review. With the exception of *Agmo*, for which a mouse conditional ko is in preparation (K. Watschinger, N. Yannoutsos, M.A. Keller, G. Golderer, G. Werner-Felmayer and E.R. Werner, unpublished work), various gene ko models exist, including *AR* [193], *PAH* [194,195], *TH* [196,197] and *TPH1/2* [198–204], as well as *NOS*s ([205] and references therein) (Table 1).

OUTLOOK AND FUTURE ISSUES

An integrated picture of BH_4 homoeostasis is still missing, linking the diverse cofactor functions and cofactor availability to the various physiological processes. In this context, we asked in our previous overview from 2000 at the end, why there are no patients with SR deficiency? Although this question was solved shortly after the review was published [1,113], the remaining 'open' problems are still 'pending'.

Below we specify issues that are of primary interest for BH_4 biology and its understanding under patho-physiological conditions:

DHPR deficiency

The most severe form of BH_4 deficiency is when DHPR is affected, which corresponds to approximately 30 % of all BH_4-deficient patients (see also http://www.biopku.org). Although these patients can be diagnosed shortly after newborn screening, clinical outcome is not always beneficial [114]. For other BH_4-deficient patients (60 % have PTPS deficiency) treatment is also possible and is more effective if diagnosed early in life. Better understanding of the pathophysiology and treatment modalities for DHPR deficiency is fervently wanted. As mentioned above, a corresponding mouse model potentially representing human DHPR deficiency was recently reported and is the basis to better investigate this disease [189].

Biopterin transport and uptake

Several investigations on transport or cellular uptake of pterins *in vitro* or *in vivo* were published by Hasegawa and co-workers [206–213]. One conclusion from their work is that, independently of the oxidation state or the stereoform (6*R* or 6*S*), the pterin seems to be converted intracellularly into 7,8-dihydrobiopterin, which is the 'universal' transport intermediate. Furthermore, inside the cell, the pterin can be fully reduced to its active 6*R*-5,6,7,8-tetrahydro (BH_4) form, thus the *6S* isomer has been converted to *6R* by a stereospecific reduction. What remains unclear is whether there is a receptor-mediated uptake, thus identification of a potential pterin transporter is of interest.

Function of BH_4 in pain

It was reported that BH_4 is an intrinsic regulator of inflammatory pain sensitivity and chronicity in animal studies, and the haplotype of the *GCH1* gene, defined by a specific SNP, identified a marker for these traits in humans [214,215]. However, others have found no association of *GCH1* genetic variations with experimental pain [216]. Additional studies are needed to confirm the association of *GCH1* polymorphisms with pain perception.

Role of BH_4 in psychiatric and neurodevelopmental disorders

Abnormal BH_4 metabolism and its potential therapeutic applications were described in a number of psychiatric and neurodevelopmental diseases such as autism spectrum disorders [217], depression [218], infantile Parkinson disease [219], Alzheimer's disease [220] or Down's syndrome [221]. Owing to its essential cofactor function for the rate-limiting hydroxylases (TH and TPHs) in the biosynthesis of catecholamines and serotonin, both dopamine and serotonin concentrations in the central nervous system may be affected. In a recent study by Schnetz-Boutaud et al. [222], 25 SNPs (single nucleotide polymorphisms) in nine genes of the BH_4 pathway in a total of 403 families were genotyped. Significant nominal association was detected in the *PTS* gene and this result was not restricted to an affected male-only subset. Multilocus interaction was detected in the BH_4 pathway alone, but not across the serotonin, dopamine and BH_4 pathways. It has been shown both in animal experiments [103] and in small clinical studies [223–225] that in some patients BH_4 can restore neurotransmitter levels in the cerebrospinal fluid and improve neurological symptoms. However, a definite role of BH_4 and/or its metabolic enzymes for these neurological abnormalities needs further understanding of cofactor homoeostasis and function in complex organisms.

Animal models for GTPCH deficiency

As mentioned above, a mouse model for the recessive form of GTPCH deficiency is not available (as it is probably embryonic lethal). Alternatively, it would be of interest to 'flox' the murine *Gch1* gene to potentially investigate tissue-specific ko mice. Furthermore, what would be the phenotype of patients with GFRP deficiency?

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DATABASES

The BIOPKU [International Database of Patients and Mutations causing BH_4-responsive hyperphenylalaninaemia/phenylketonuria (HPA/PKU)], BIODEF (International Database of Tetrahydrobiopterin Deficiencies) and BIOMDB (Database of Mutations Causing Pediatric neurotransmitter diseases) databases are available at http://www.biopku.org/.

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